21.11.23

OPENCAMPUS.sh

Einführung in Data Science und maschinelles Lernen

VERSIONIERUNG MIT GIT (TEIL 2) UND START DES AUSWERTUNGSPROJEKTES

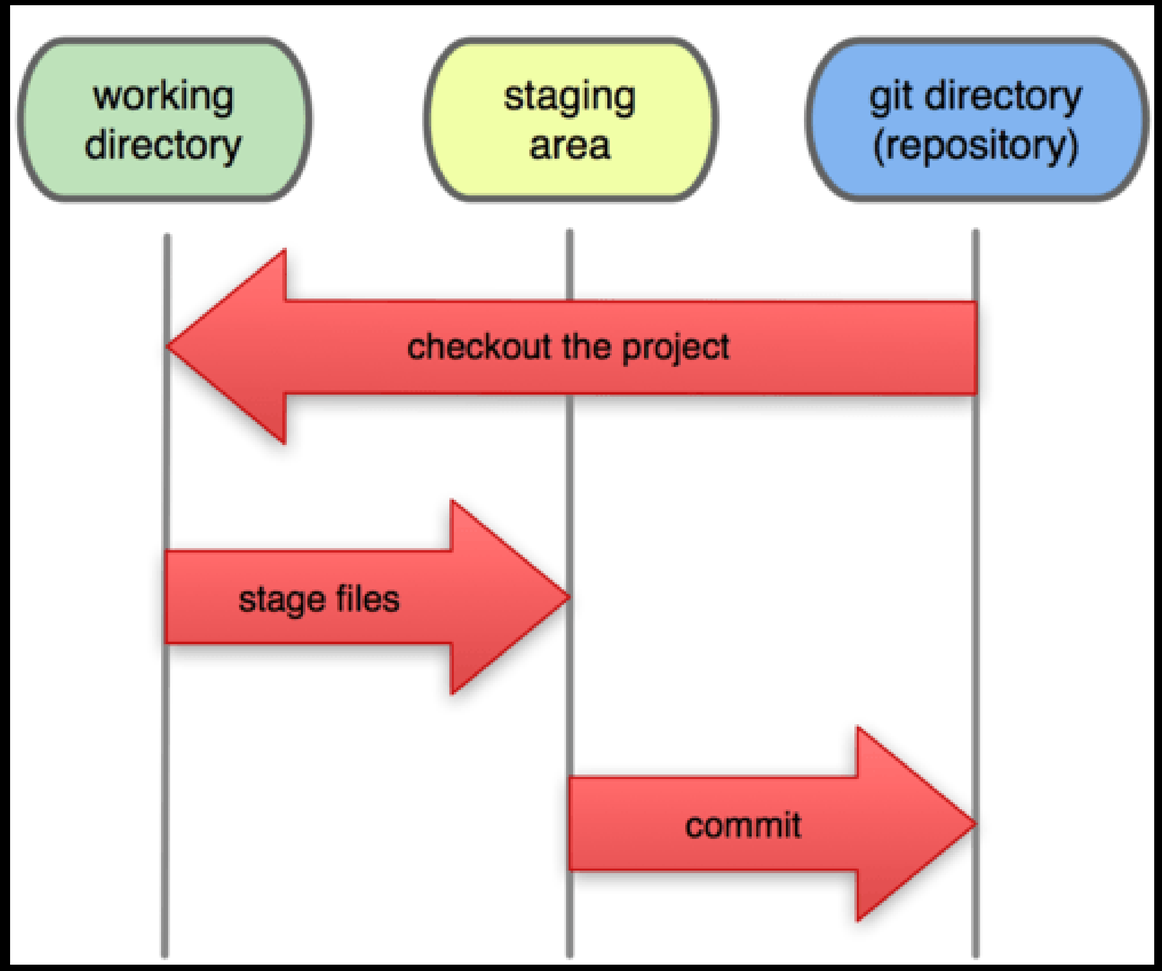
- Ergänzung der Teams
- Besprechung der Aufgaben
- Einführung in Git Teil 2
- Thematische Einführung zum Thema KI bzw. maschinelles Lernen

ERGÄNZUNG DER TEAMS

BREAKOUT

- Bestimmt eine Person bei Euch im Team, die im Anschluss an die Breakout-Session ein GitHub-Repository unter seinem Account anlegen soll.
- Besprechung der Aufgaben

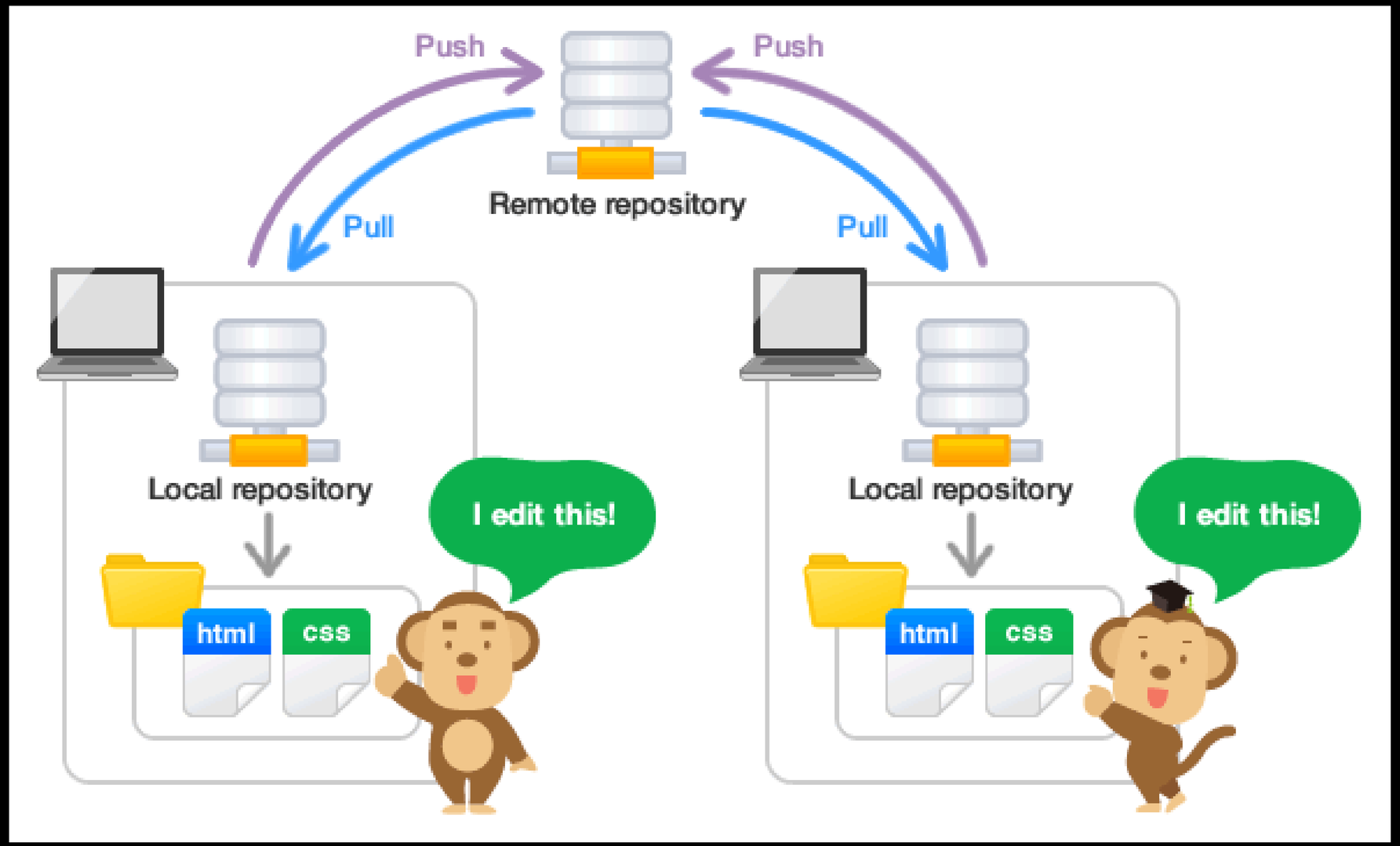
VERSIONIERUNG MIT GIT



Eine Datei kann drei mögliche Zustände haben:

- **modified („geändert“)**
- **staged („vorgemerkt“) und**
- **committed („versioniert“).**

LOKALE UND REMOTE REPOSITORIES



AUTHENTIFIZIERUNG

- Du bzw. die Anwendung auf Deinem Rechner benötigt Lese- und Schreibrechte auf dem Remote-Server.
- Dazu musst Du beim Lesen und Schreiben einen persönlichen Schlüssel verwenden, den nur Du und der Remote-Server kennen.
- Über diesen Schlüssel kann der Remote-Server Dich und Deine Rechte auf dem Server identifizieren.

AUTHENTIFIZIERUNG BEI GITHUB

Schritt 1:
Du erstellst Dir bzw. RStudio einen persönlichen Schlüssel.

- **Gehe im RStudio Menu zu**
  Tools → Global Options → Git SVN
- **Klicke dort „*Create SSH Key…*" (falls noch kein Schlüssel generiert wurde).**
  - **Klicke dort auf „*Create*" (Passwort brauchst Du nicht vergeben) und**
  - **im folgenden Fenster auf „Close"**
- **Klicke dann auf "*View public key*" und kopiere den angezeigten Schlüssel in die Zwischenablage.**

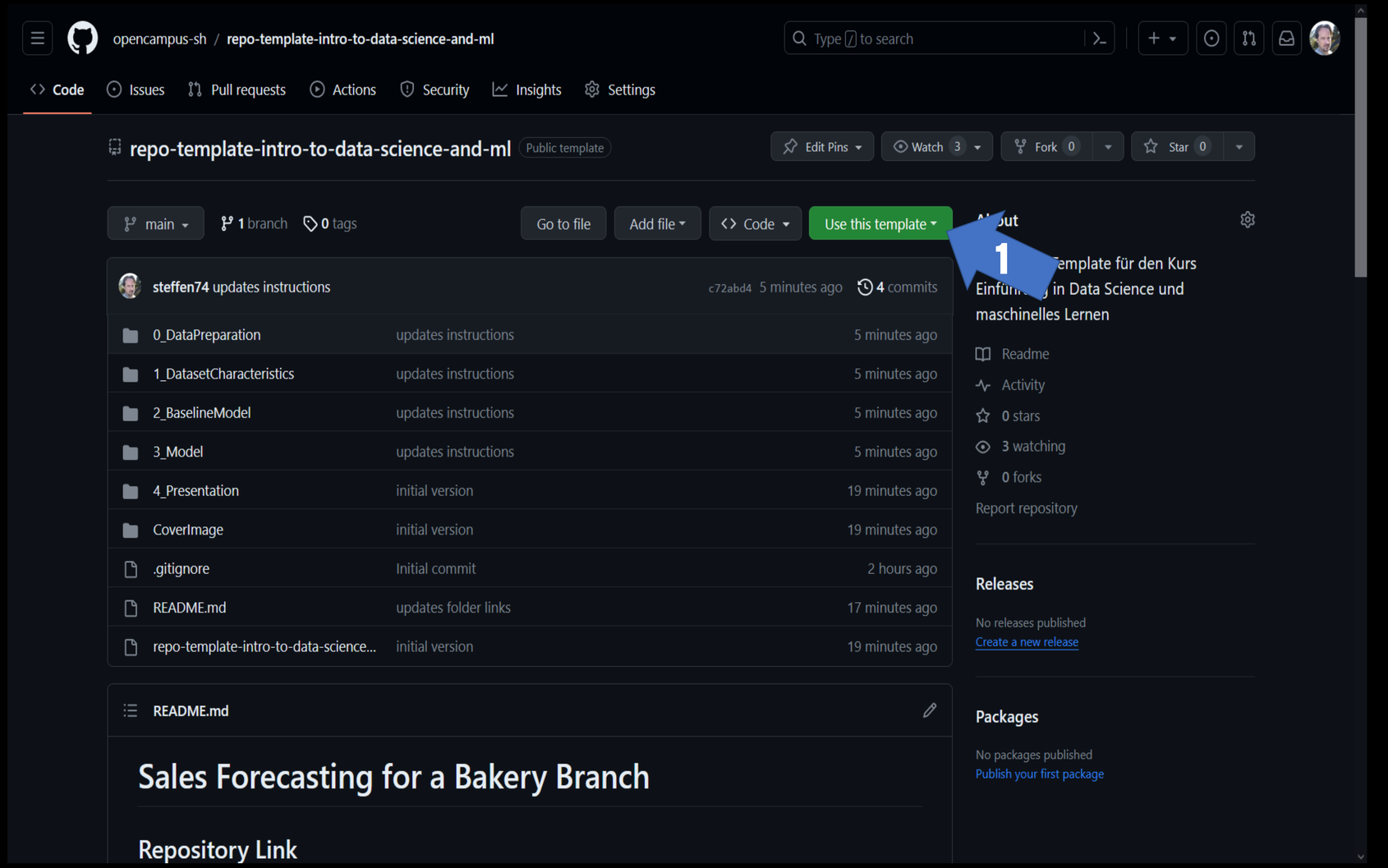
Schritt 2:
Du speicherst diesen Schlüssel bei GitHub in Deinem Profil, damit GitHub Dich mit diesem Schlüssel identifizieren kann.

- **Klicke in GitHub oben rechts auf Dein Profilbild und gehe zu „Settings".**
- **Klicke links im Menu auf „SSH und GPG Keys".**
- **Klicke auf „New SSH key" und füge dort den Schlüssel aus der Zwischenablage ein.**

REMOTE REPOSITORY ANLEGEN

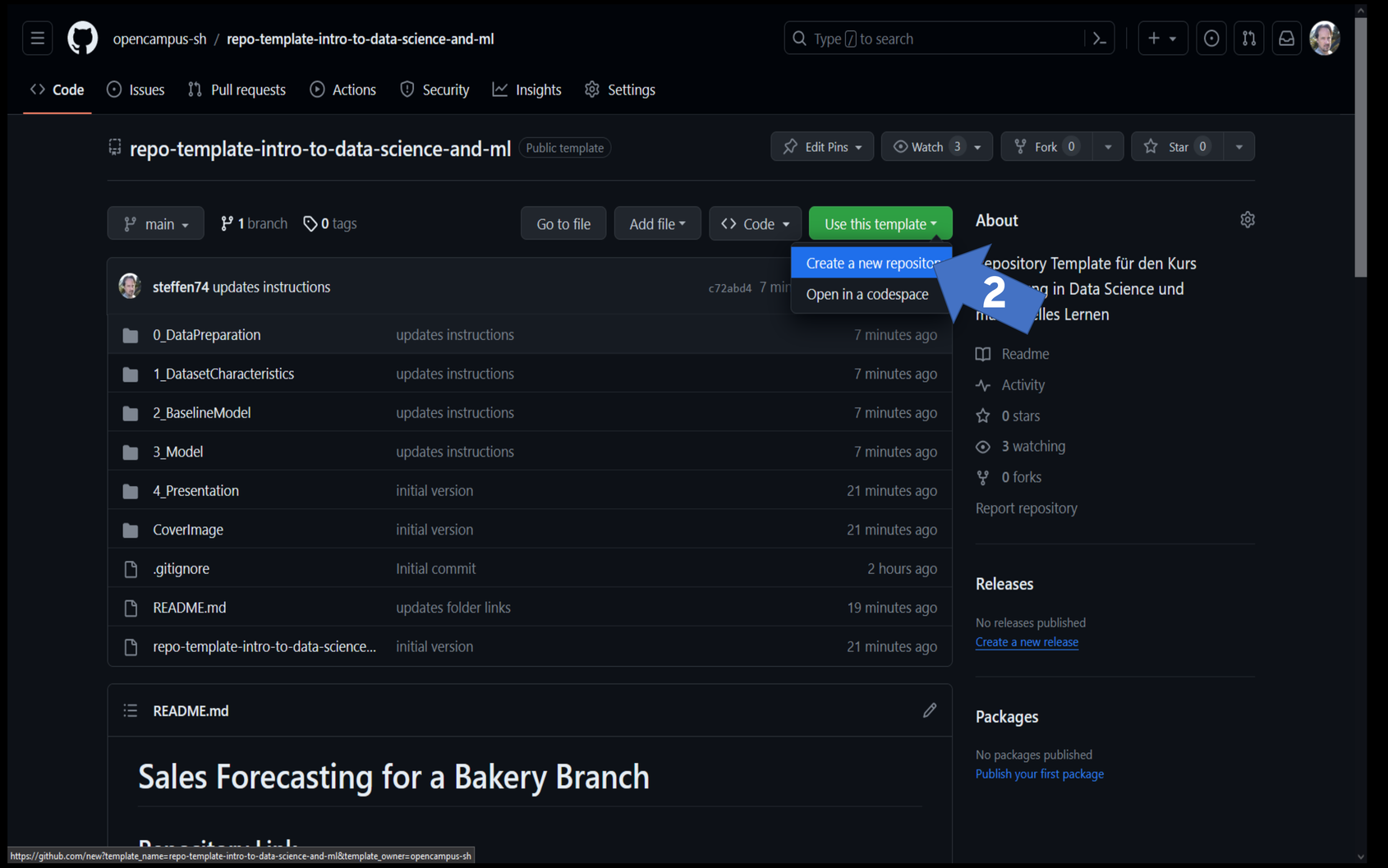
- Repository auf GitHub anlegen
- Andere GitHub-User die Lese- oder Schreibberechtigungen haben sollen, als „Collaborator“ zum Repository hinzufügen.

PROJEKT-REPOSITORY ANLEGEN



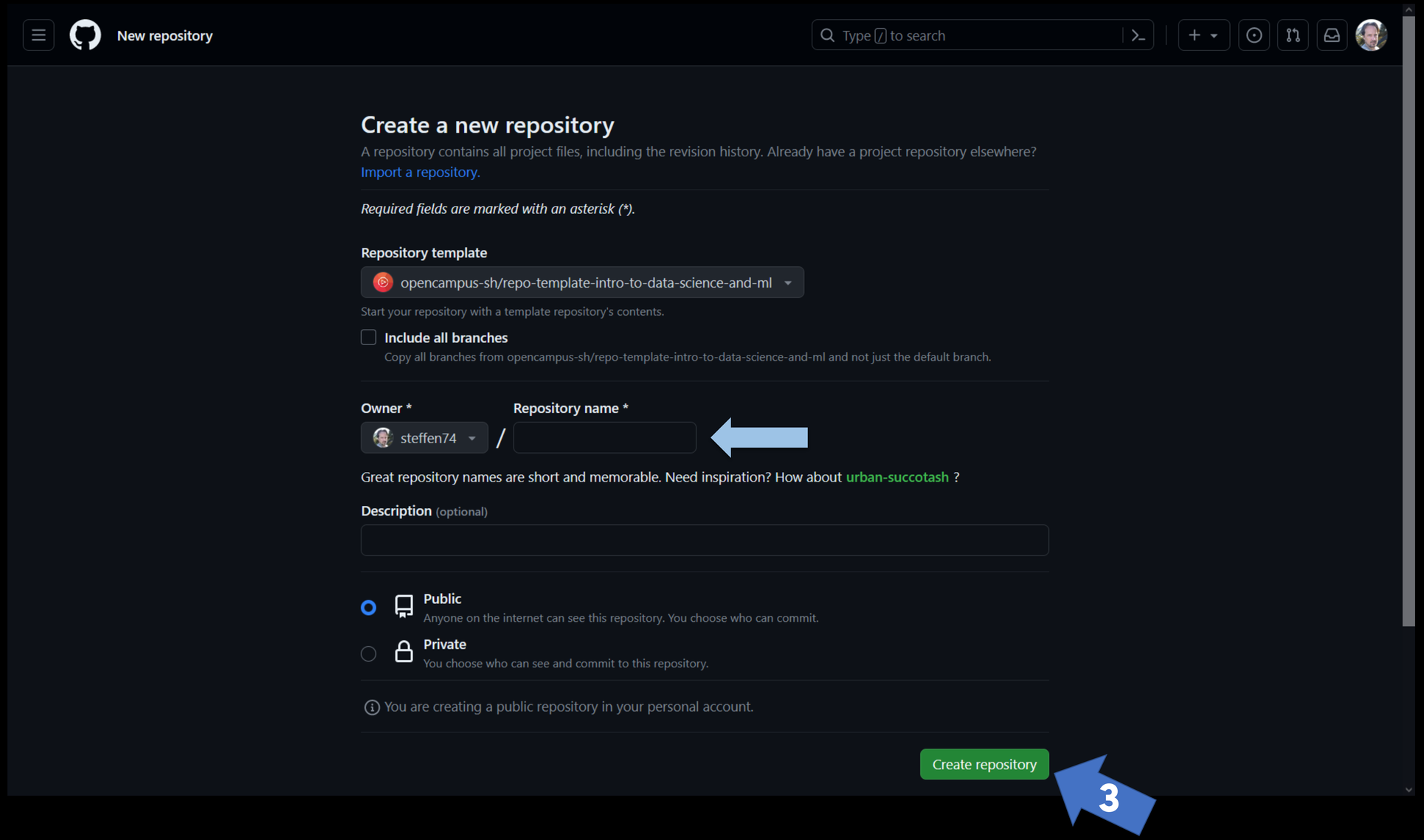
https://github.com/opencampus-sh/repo-template-intro-to-data-science-and-ml/tree/main

PROJEKT-REPOSITORY ANLEGEN

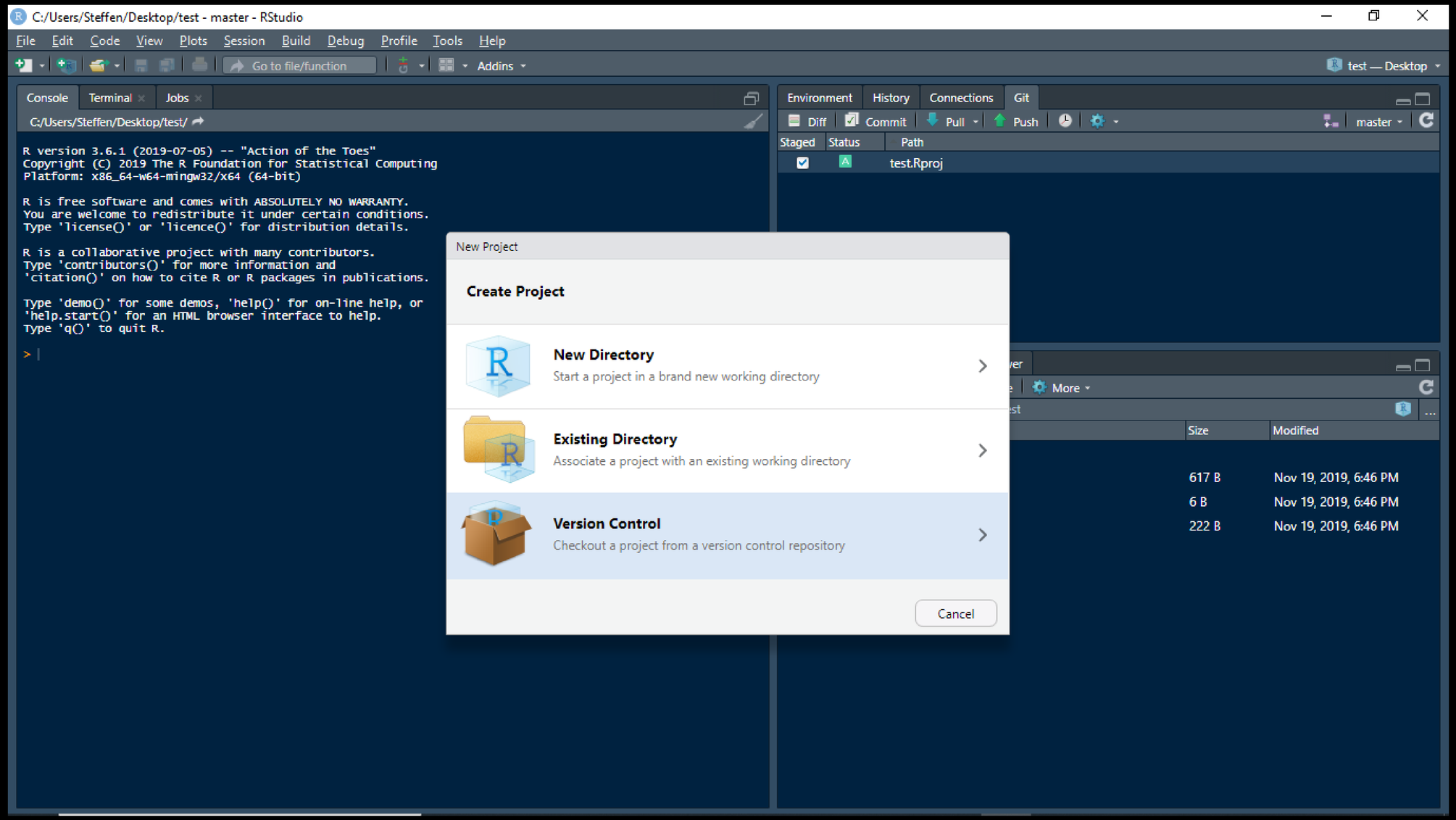


https://github.com/opencampus-sh/repo-template-intro-to-data-science-and-ml/tree/main

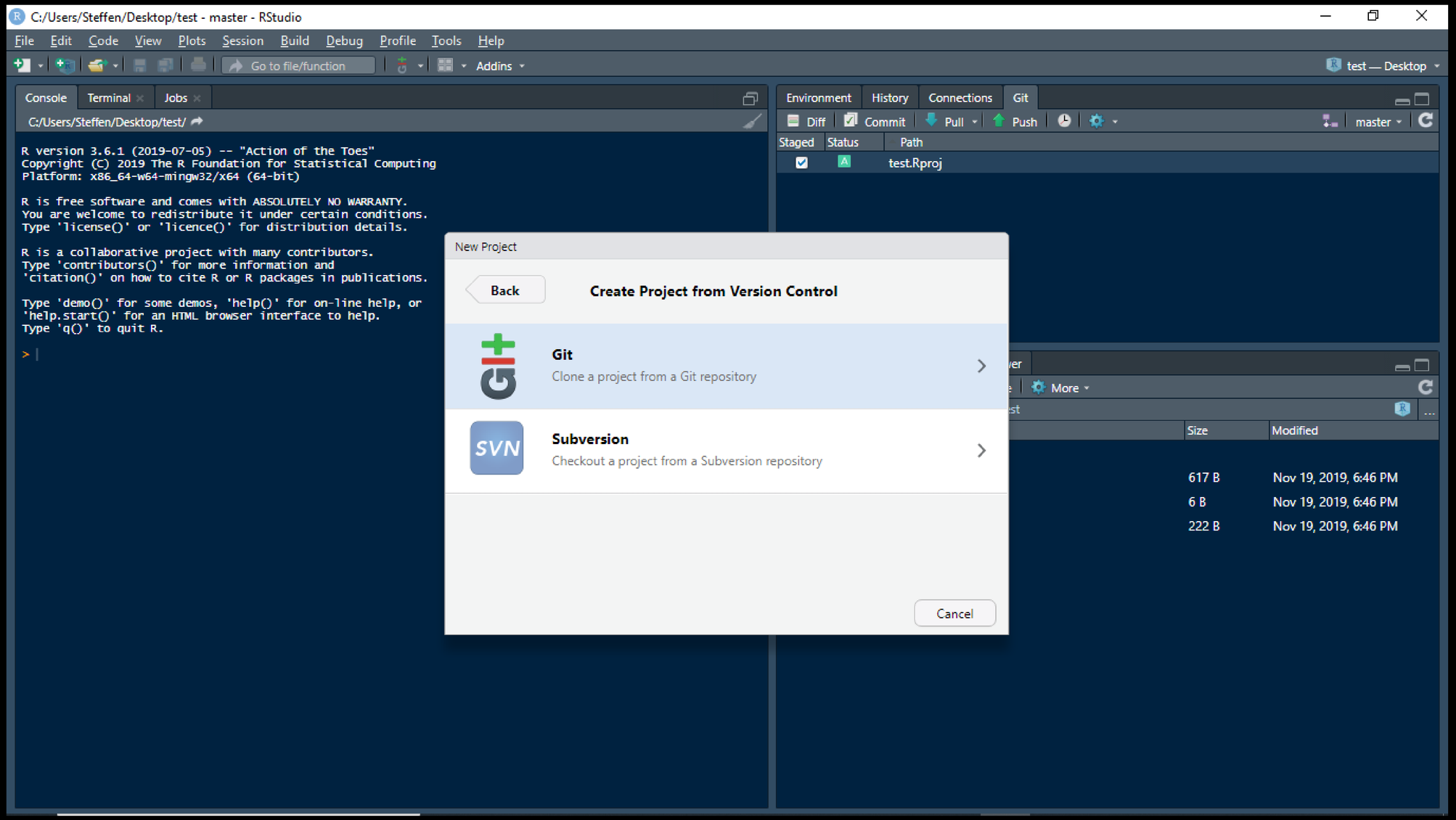
PROJEKT-REPOSITORY ANLEGEN



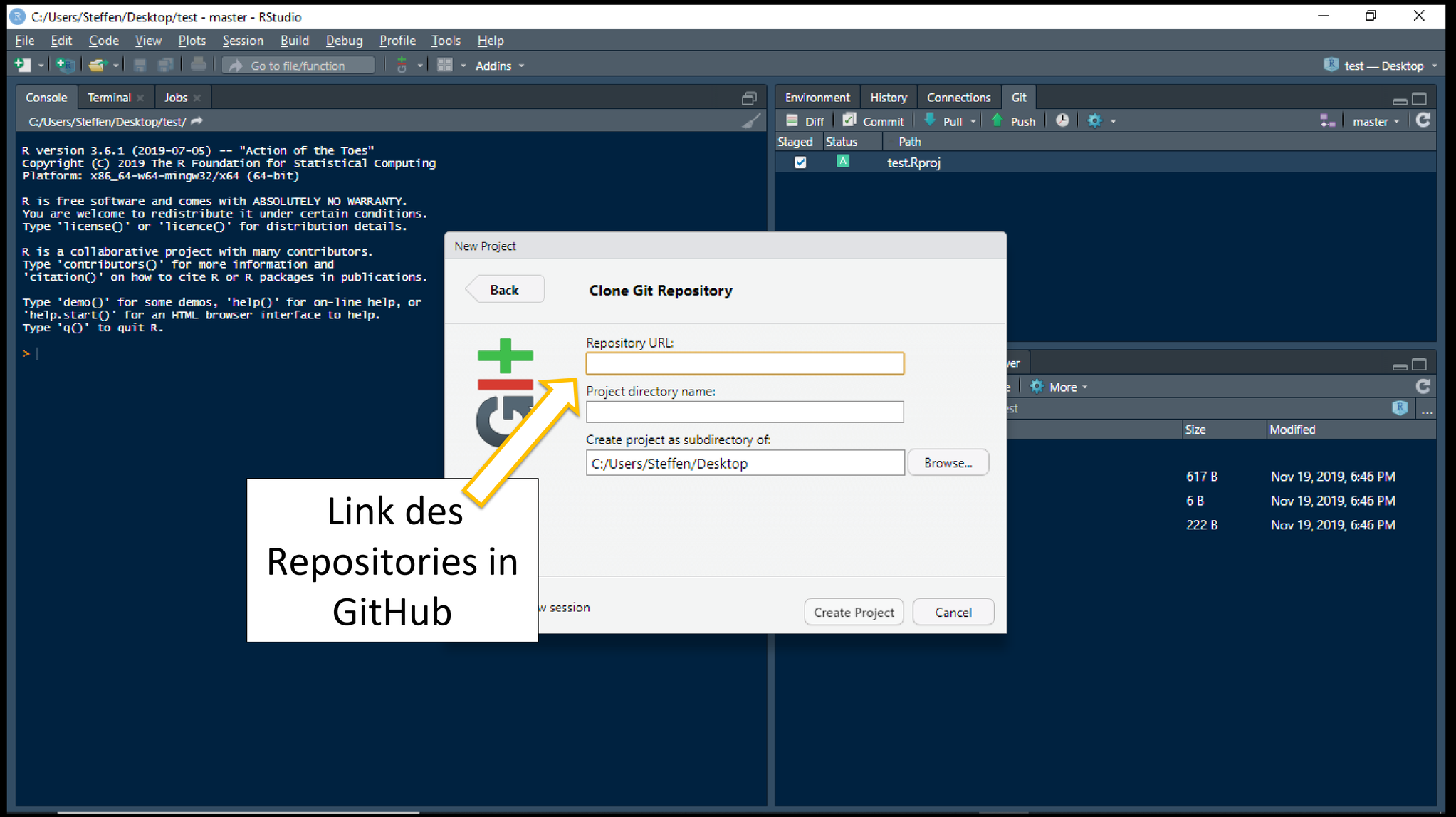
GITHUB REPOSITORY ALS PROJEKT IMPORTIEREN (1)



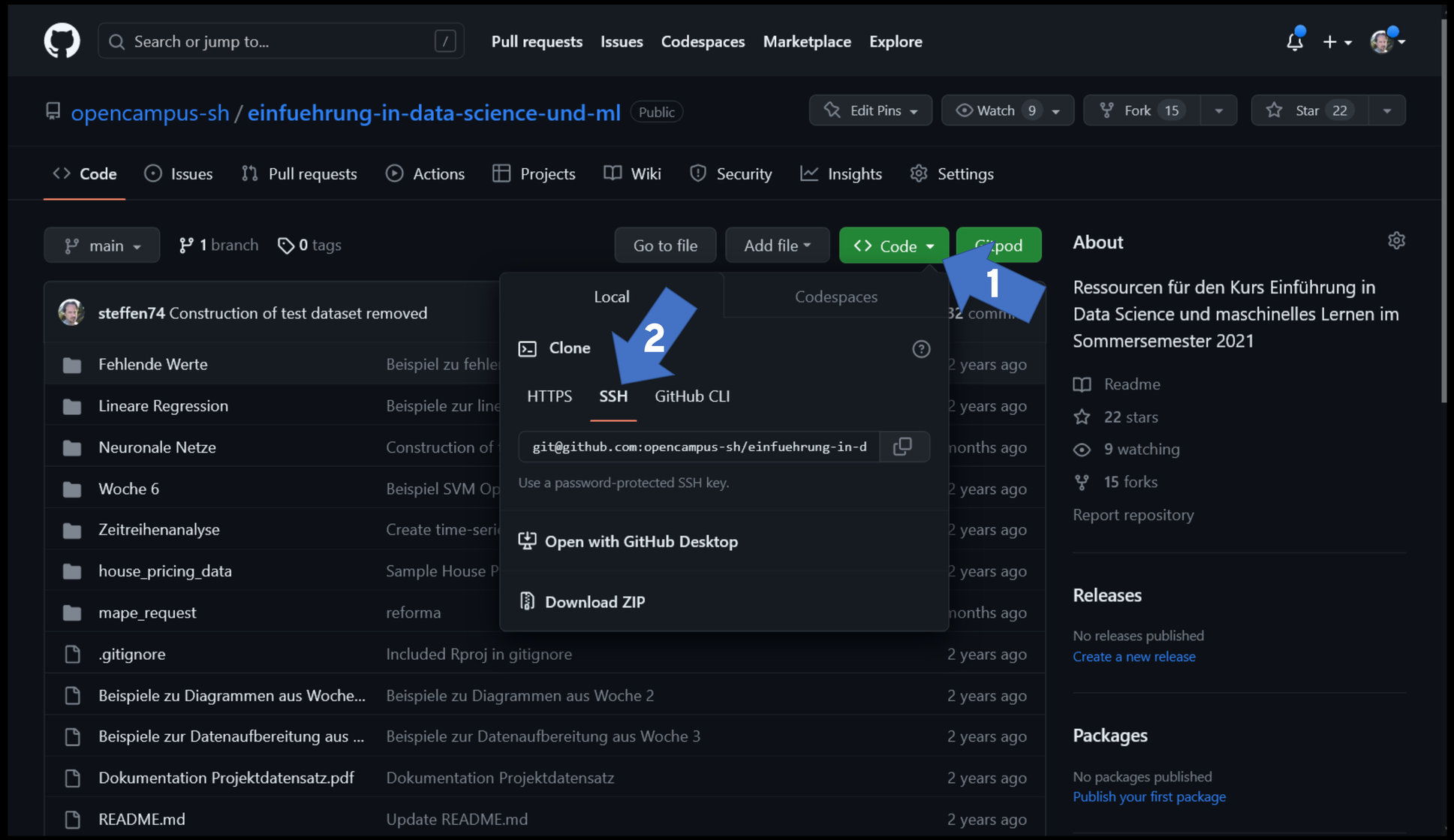
GITHUB REPOSITORY ALS PROJEKT IMPORTIEREN (2)



GITHUB REPOSITORY ALS PROJEKT IMPORTIEREN (3)



LINK DES REPOSITORIES IN GITHUB



PROJEKT

Präsentation

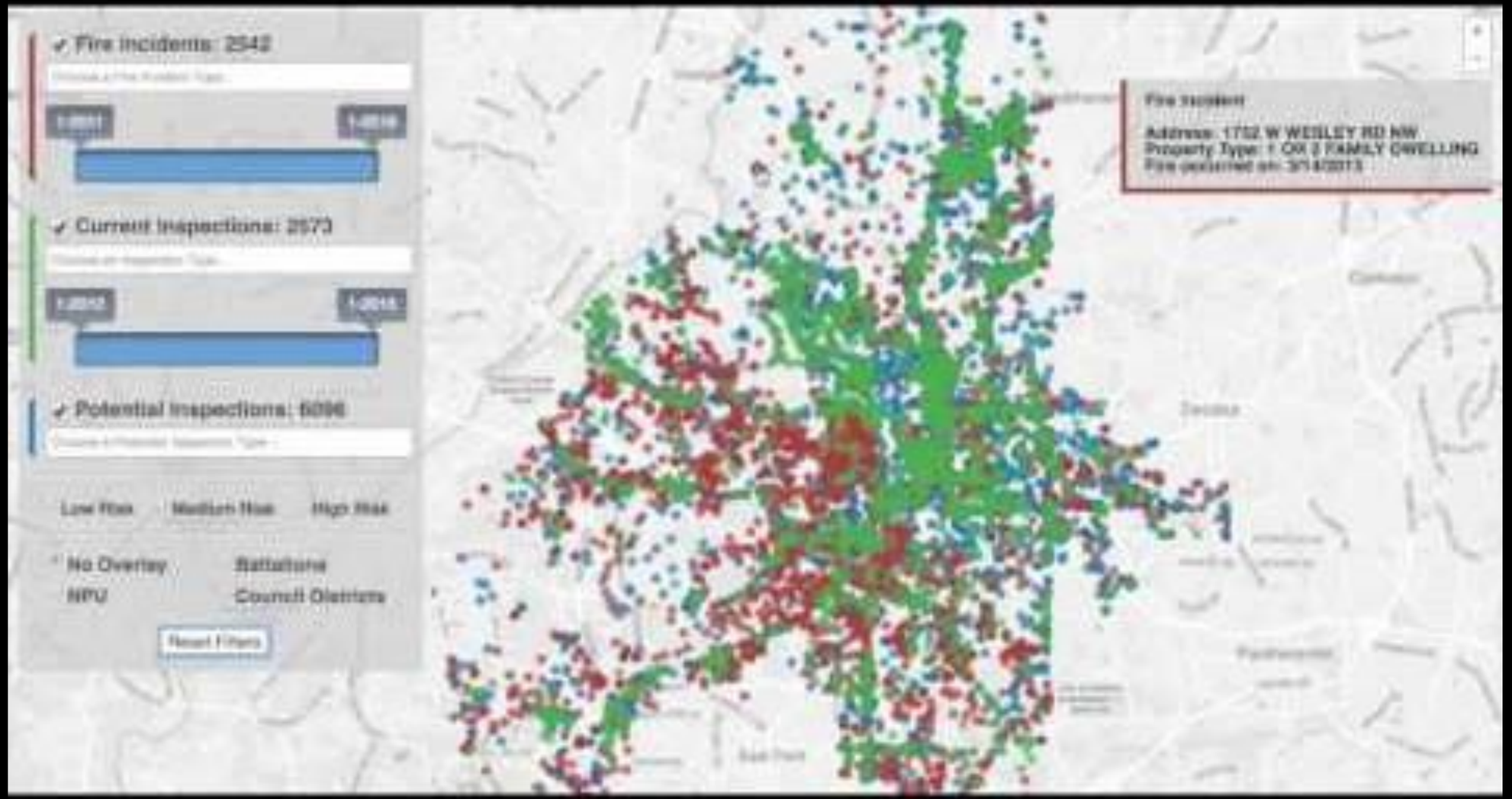
- Eure Namen im Titelteil
- Vorstellung der selbst erstellten Variablen
- Balkendiagramme mit Konfidenzintervallen für zwei selbst erstellte Variablen
- Optimierung eines linearen Modells („Baseline Modell“):
  (a) Modellgleichung
  (b) Adjusted r^2
  (c) MAPE für den Zeitraum vom 1.8.2018 bis 30.7.2019
- Optimierung eines neuronalen Netzes:
  (a) Source Code der Definition des neuronalen Netzes
  (b) Darstellung der Loss-Funktionen für Trainings- und Validierungsdatensatz
  (c) MAPE je Warengruppe für den Zeitraum vom 1.8.2018 bis 30.7.2019

Repository

- Datensatz
- Programmcode
- Präsentation

AKTUELLE ANWENDUNGEN IM KI- BEREICH

RISIKOPROGNOSEN



SCHACHCOMPUTER



AlphaGo – The Movie:
https://www.youtube.com/watch?v=WXuK6gekU1Y

ROBOTERSTEUERUNG



BILD- UND VIDEOGENERIERUNG

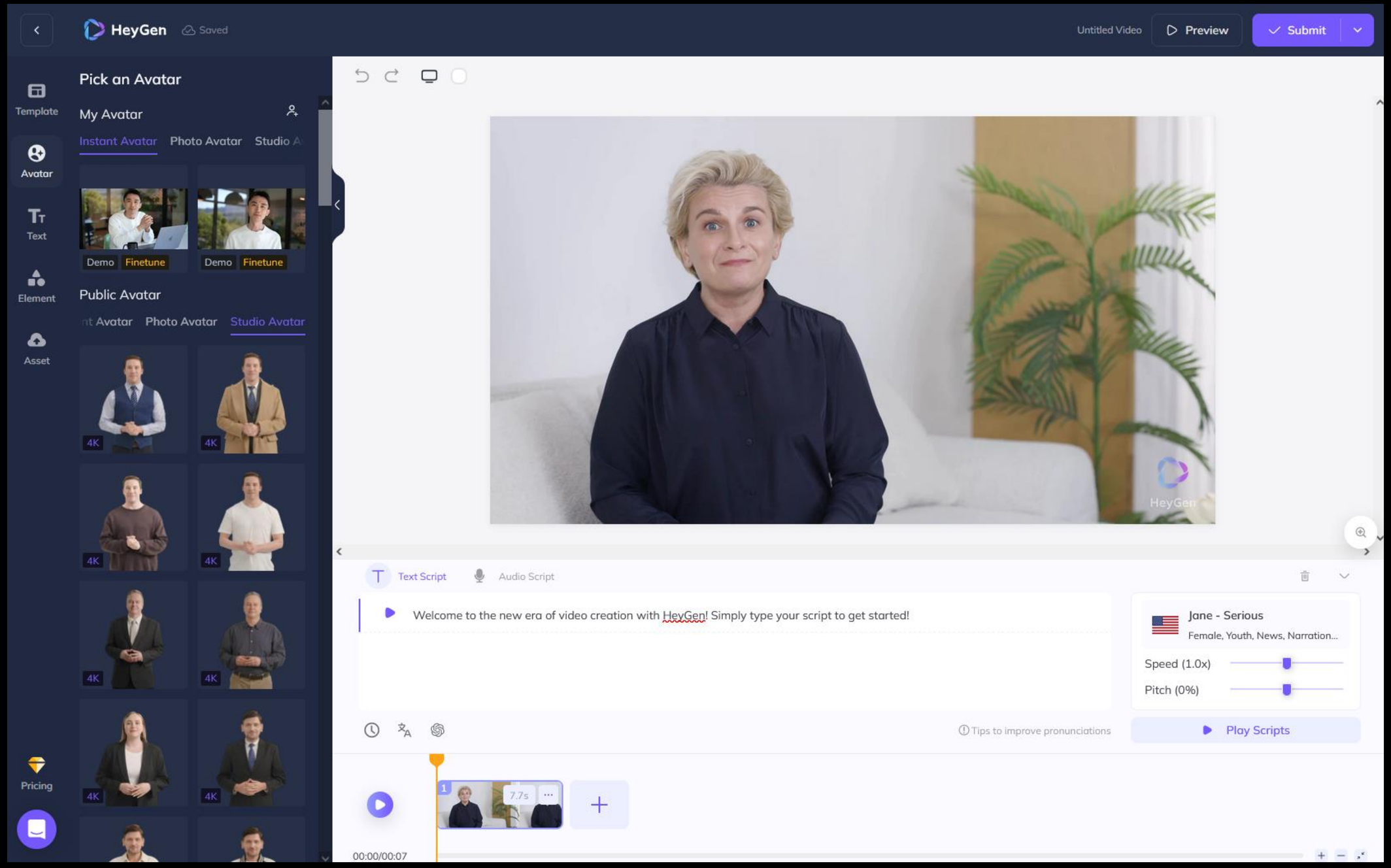




Advay Pal [@advaypal]. (2023, April 26). what have we done? Https://t.co/DVC0zL2Xgh [Tweet]. Retrieved October 9, 2023, from Twitter website: https://twitter.com/advaypal/status/1651355572980310016



HeyGen—AI Spokesperson Video Creator. (2023). Retrieved October 9, 2023, from https://app.heygen.com/home


HeyGen
Saved
Untitled Video
Preview
Submit
Template
Avatar
Text
Element
Asset
Pricing
Pick an Avatar
My Avatar
Instant Avatar
Photo Avatar
Studio A
Demo
Finetune
Demo
Finetune
Public Avatar
nt Avatar
Photo Avatar
Studio Avatar
4K
4K
4K
4K
4K
4K
4K
4K
HeyGen
Text Script
Audio Script
Welcome to the new era of video creation with HeyGen! Simply type your script to get started!
Jane - Serious
Female, Youth, News, Narration...
Speed (1.0x)
Pitch (0%)
Tips to improve pronunciations
Play Scripts
1
7.7s
00:00/00:07

NATURAL LANGUAGE PROCESSING

GPT-3

Explain Kants Categorial Imperative

Share Try again

The Categorial Imperative is an ethical theory proposed by the German philosopher Immanuel Kant. This was one of two main moral theories in his philosophy (the other being Deontology). His 'categorical imperative' proposes that an action can be considered moral if it could potentially become a universal law.

The Categorial Imperative is essentially the idea that the ends don't justify the means. Only actions which are consistent with a universal law (that can be applied consistently to all human beings) can be considered moral.

The Categorial Imperative also suggests that the motivations behind an action are irrelevant. For example, if I was to kill my enemy in cold blood, but did it out of compassion and love for humanity, this would not make it moral - as the act itself is contradicting a universal law.

However, it is important to note that the Categorial Imperative also says that an action can be considered moral if there are no other means of achieving a goal. For example, if I was in need of money for my own survival and I needed to steal it from someone else - without causing any harm.

Overall, I would say Kant's Categorial Imperative is a moral theory that puts great emphasis on the principle of 'doing unto others as you would have them do to you'.

It is also worth noting that Kant's Categorial Imperative disagrees with the Utilitarian theory of ethics, as it puts more emphasis on an action being consistent with a universal law rather than providing the greatest good to the greatest number.

Schreiner, M. (2020, August 31). GPT-3: Diese fünf Apps zeigen das KI-Potenzial [Blog]. Retrieved May 4, 2021, from MIXED website: https://mixed.de/gpt-3-diese-fuenf-apps-zeigen-das-ki-potenzial/

CHATBOTS ALS AGENTEN

ChatGPT Plugins

How can I help you today?

Geben Sie mir Ideen
about how to plan my New Years resolutions

Zeige mir einen Code-Ausschnitt
eines Sticky-Headers einer Website

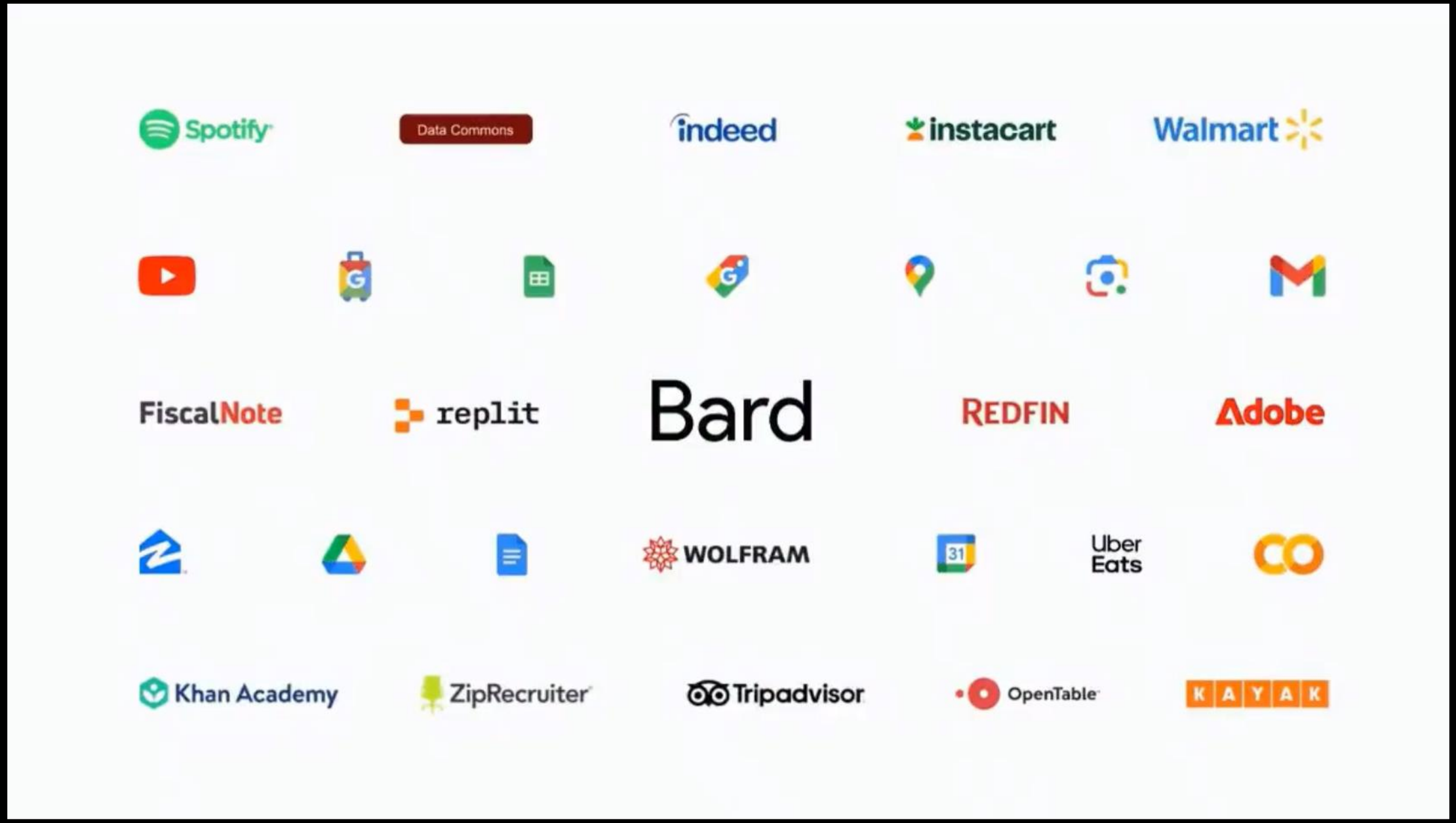
Erfinde eine Geschichte
über Sharky einen Zahnputzenden Hai-Superh...

Konzepte entwickeln
Für ein Retro-Arkaden-Spiel

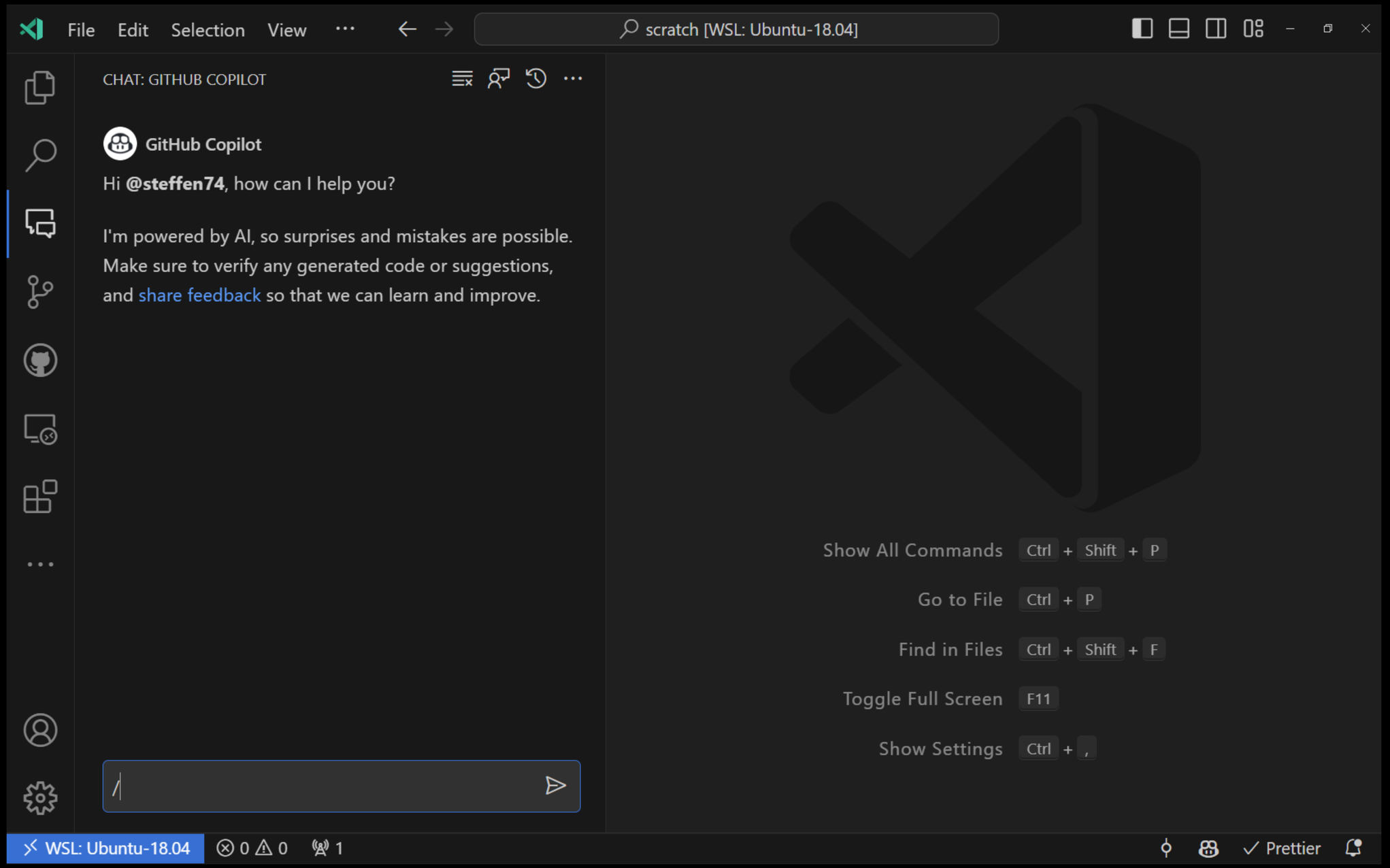
Message ChatGPT...

ChatGPT can make mistakes. Consider checking important information.

?


Spotify
Data Commons
indeed
instacart
Walmart
FiscalNote
replit
Bard
REDFIN
Adobe
WOLFRAM
Uber Eats
Khan Academy
ZipRecruiter
Tripadvisor
OpenTable
KAYAK

CHATBOTS ZUR PROGRAMMIERUNG


File
Edit
Selection
View
scratch [WSL: Ubuntu-18.04]
CHAT: GITHUB COPILOT
GitHub Copilot
Hi @steffen74, how can I help you?
I'm powered by AI, so surprises and mistakes are possible.
Make sure to verify any generated code or suggestions,
and share feedback so that we can learn and improve.
/
Show All Commands Ctrl + Shift + P
Go to File Ctrl + P
Find in Files Ctrl + Shift + F
Toggle Full Screen F11
Show Settings Ctrl + ,
WSL: Ubuntu-18.04
0 0
1
Prettier

Merged **Adds New Tablegrid generic component** #809

ericstumper merged 14 commits into `staging` from `develop` 19 hours ago

review-agent-prime bot commented 20 hours ago

frontend-nx/apps/edu-hub/queries/generated /DegreeParticipantsWithDegreeEnrollments.ts

Consider adding comments to the newly added code to improve readability and maintainability. This will help other developers understand the purpose of the code and any important details they should be aware of.

Create Issue

```
// This is an example of how you might comment the new code
// URL to the file containing the user's attendance certificate (if he obtained one)
attendanceCertificateURL: string | null;
```

frontend-nx/apps/edu-hub/queries/user.ts

Consider optimizing the query by only fetching the necessary fields. This will reduce the amount of data transferred and improve the performance of the application.

Create Issue

```
// This is an example of how you might optimize the query
User(where: { status: { _eq: ACTIVE } }) {
  id
  firstName
  lastName
}
```

Consider sanitizing the input to the query to prevent SQL injection attacks. This can be done by using parameterized queries or prepared statements.

Create Issue

```
// This is an example of how you might sanitize the input
query UserForSelection1($searchValue: String!) {
  User(where: { firstName: { _ilike: $searchValue } }) {
    id
    firstName
    lastName
```

MULTIMODALE MODELLE



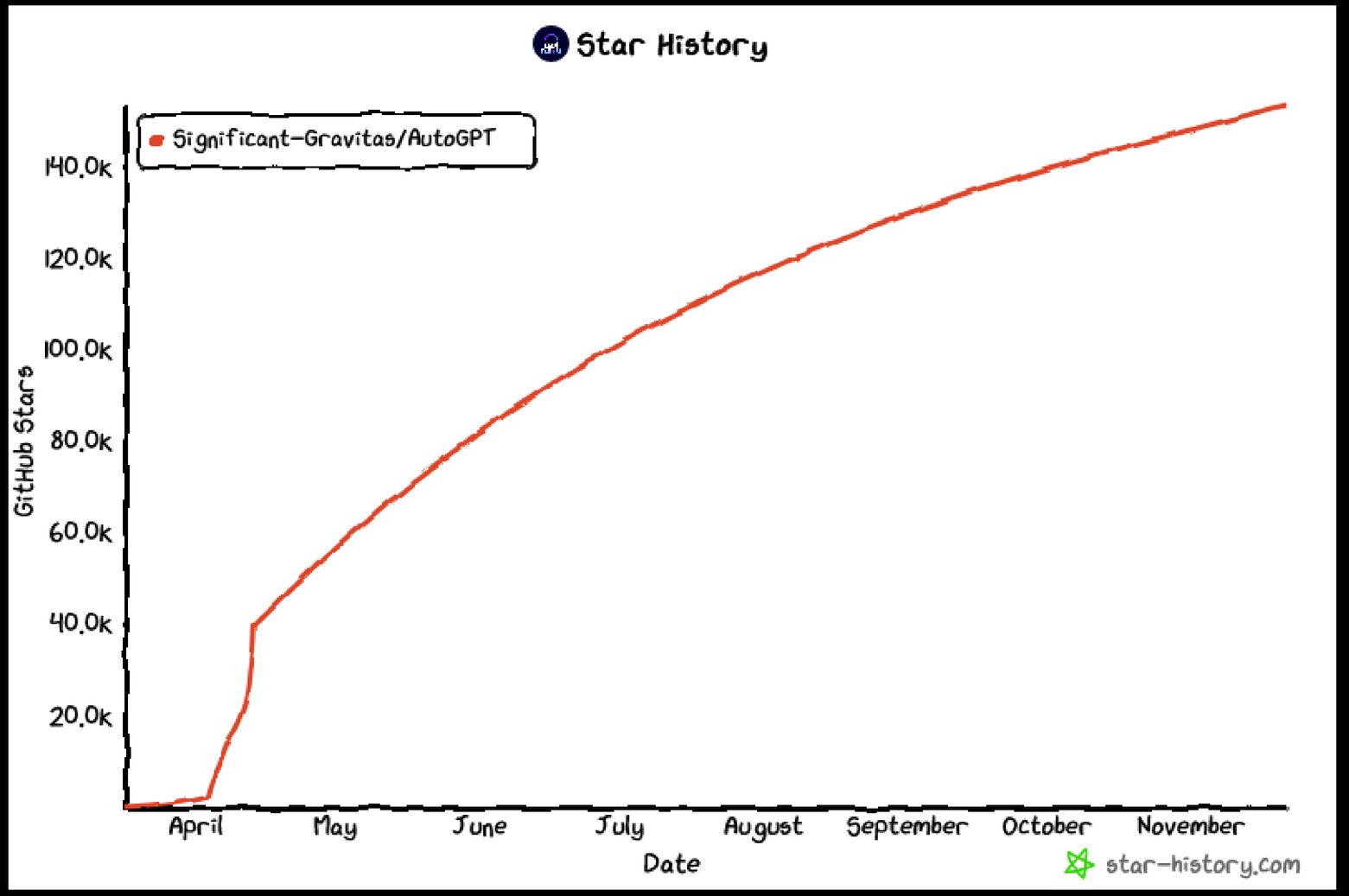
KREATIVITÄT

“A BOWL OF SOUP THAT LOOKS LIKE A MONSTER KNITTED OUT OF WOOL.”

“TEDDY BEARS MIXING SPARKLING CHEMICALS AS MAD SCIENTISTS IN A STEAMPUNK STYLE.”

OPEN SOURCE KI

AUTO GPT

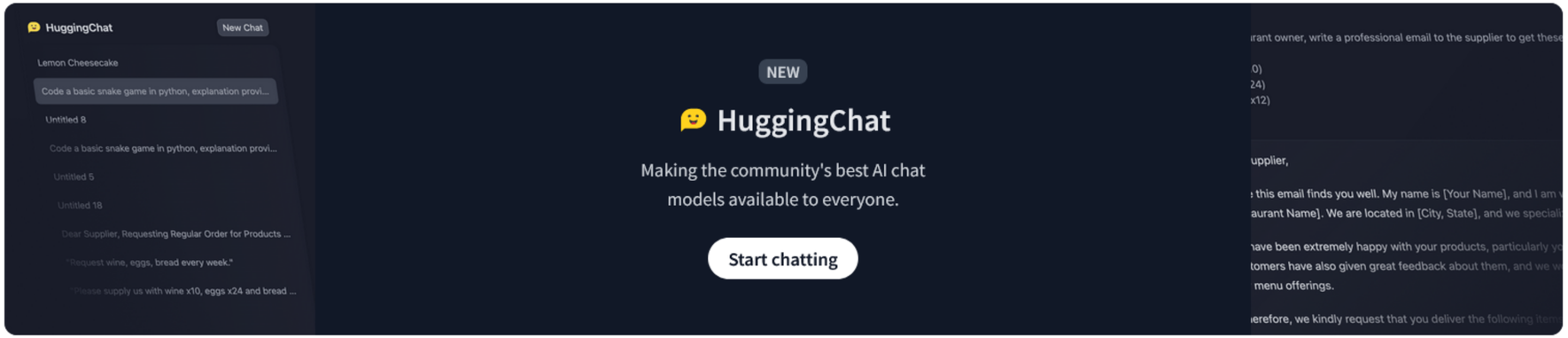

Star History
Significant-Gravitas/AutoGPT
140.0k
120.0k
100.0k
80.0k
60.0k
40.0k
20.0k
GitHub Stars
April
May
June
July
August
September
October
November
Date
star-history.com

AUTOGPT

🚀 Features

- 🌐 Internet access for searches and information gathering
- 💾 Long-term and short-term memory management
- 🧠 GPT-4 instances for text generation
- 🔗 Access to popular websites and platforms
- 🗃️ File storage and summarization with GPT-3.5
- 🔌 Extensibility with Plugins

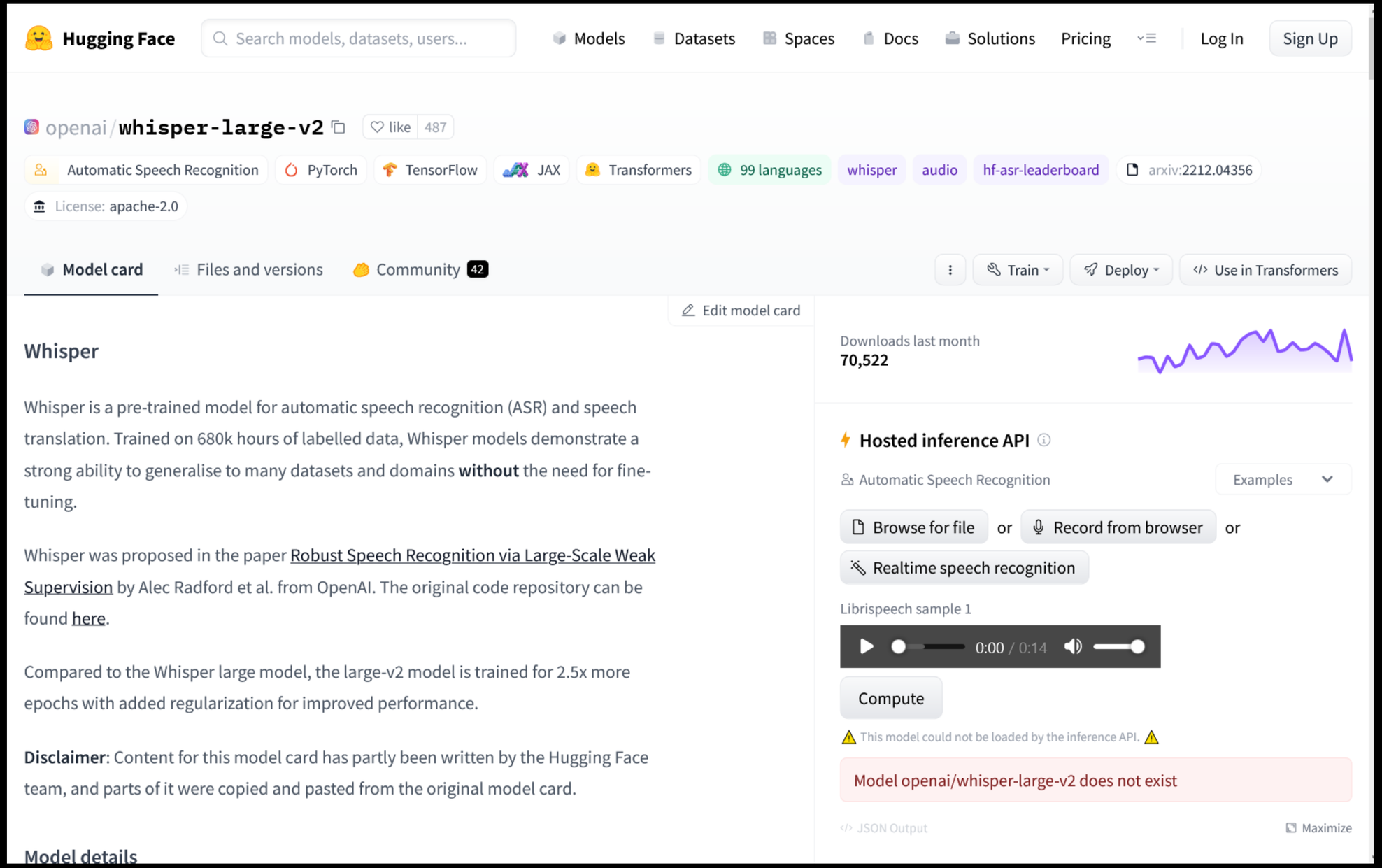
Hugging Face | Search models, datasets, users... | Models | Datasets | Spaces | Docs | Solutions | Pricing | Log In | Sign Up

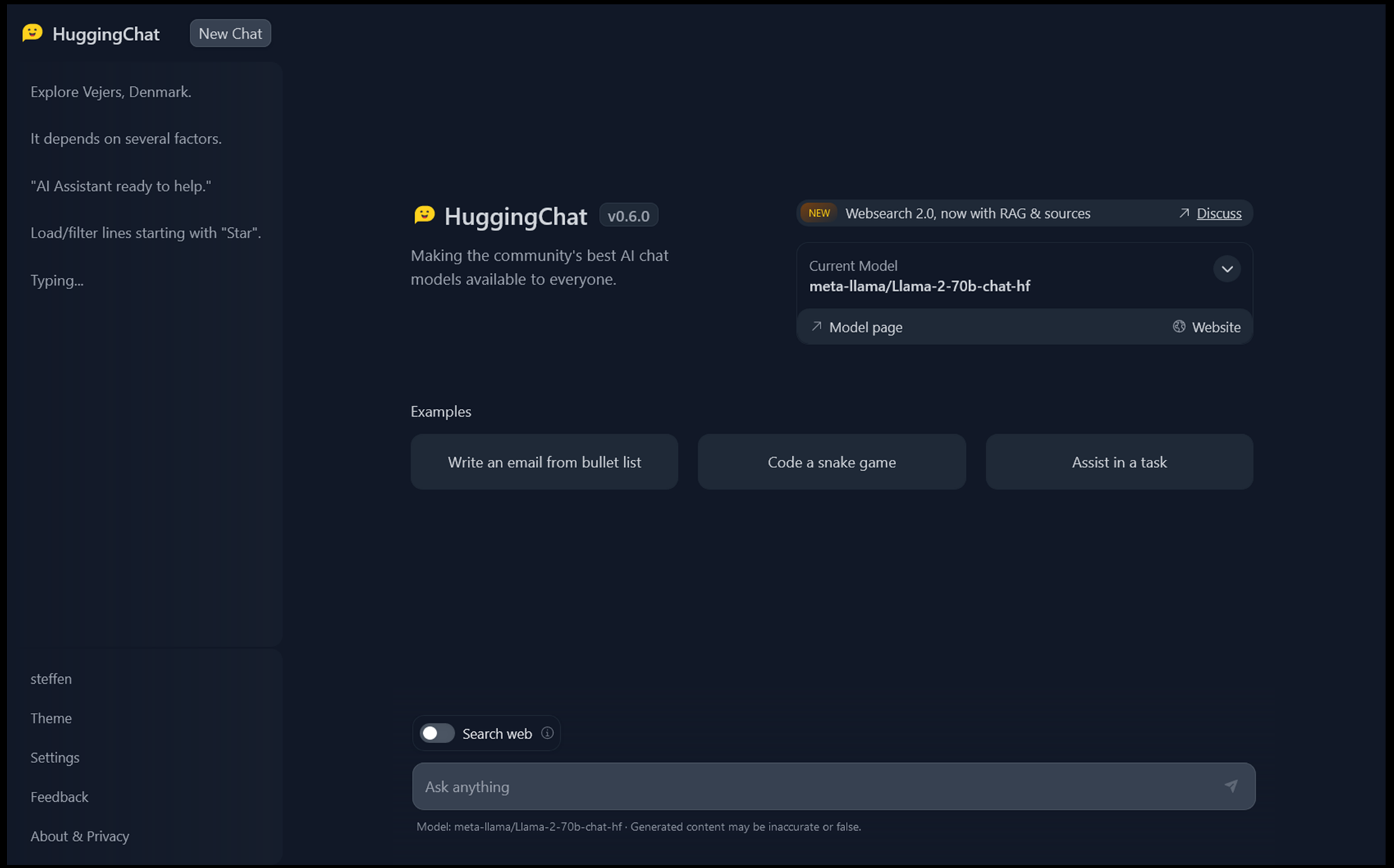


The AI community building the future.

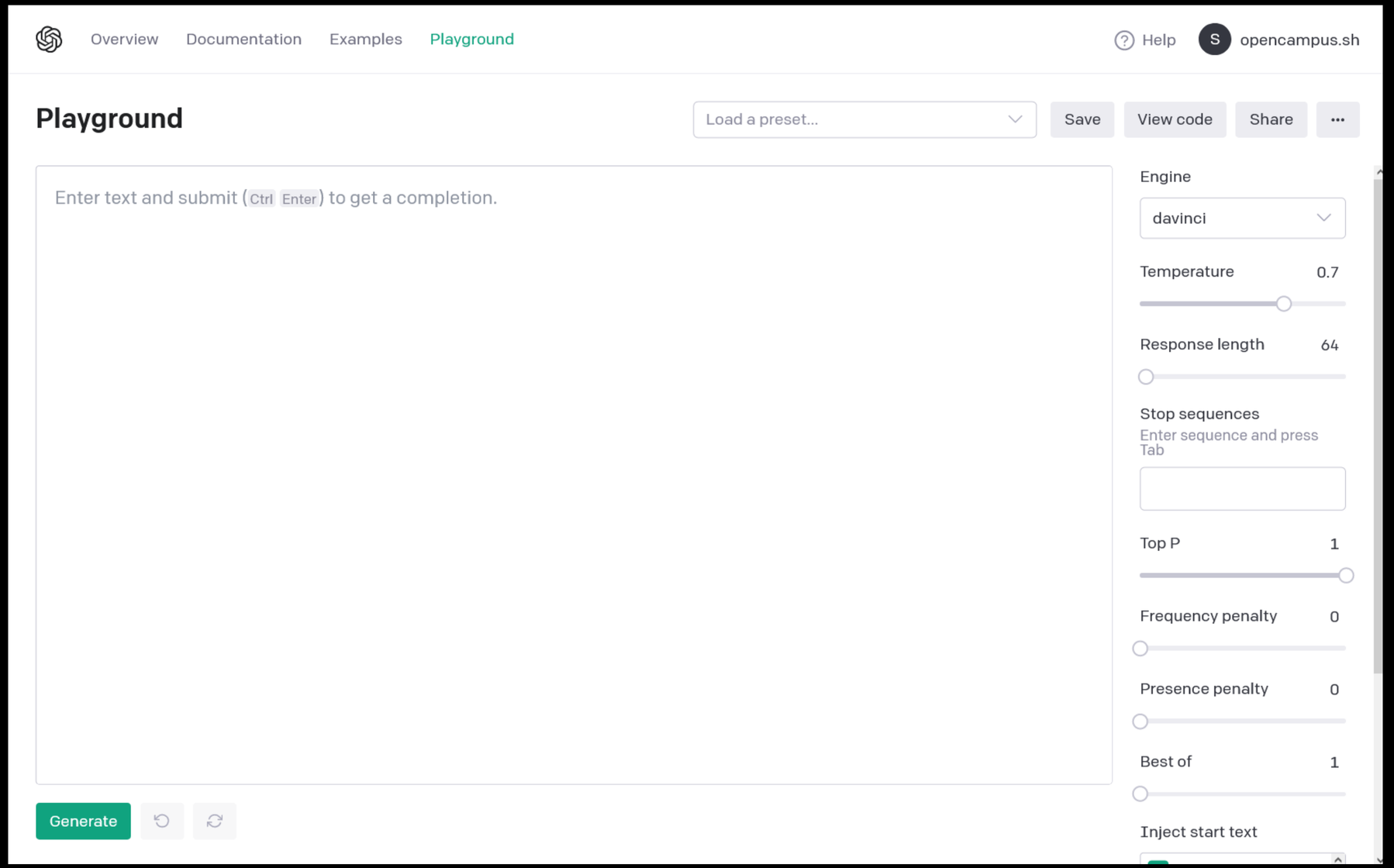
Build, train and deploy state of the art models powered by the reference open source in machine learning.

Star 98,534


Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Docs
Solutions
Pricing
Log In
Sign Up
openai / whisper-large-v2
like 487
Automatic Speech Recognition
PyTorch
TensorFlow
JAX
Transformers
99 languages
whisper
audio
hf-asr-leaderboard
arxiv:2212.04356
License: apache-2.0
Model card
Files and versions
Community 42
Train
Deploy
Use in Transformers
Edit model card
Whisper
Whisper is a pre-trained model for automatic speech recognition (ASR) and speech translation. Trained on 680k hours of labelled data, Whisper models demonstrate a strong ability to generalise to many datasets and domains without the need for fine-tuning.
Whisper was proposed in the paper Robust Speech Recognition via Large-Scale Weak Supervision by Alec Radford et al. from OpenAI. The original code repository can be found here.
Compared to the Whisper large model, the large-v2 model is trained for 2.5x more epochs with added regularization for improved performance.
Disclaimer: Content for this model card has partly been written by the Hugging Face team, and parts of it were copied and pasted from the original model card.
Model details
Downloads last month
70,522
Hosted inference API
Automatic Speech Recognition
Examples
Browse for file
or
Record from browser
or
Realtime speech recognition
Librispeech sample 1
0:00 / 0:14
Compute
This model could not be loaded by the inference API.
Model openai/whisper-large-v2 does not exist
JSON Output
Maximize


HuggingChat
New Chat
Explore Vejers, Denmark.
It depends on several factors.
"AI Assistant ready to help."
Load/filter lines starting with "Star".
Typing...
steffen
Theme
Settings
Feedback
About & Privacy
HuggingChat
v0.6.0
Making the community's best AI chat models available to everyone.
NEW
Websearch 2.0, now with RAG & sources
Discuss
Current Model
meta-llama/Llama-2-70b-chat-hf
Model page
Website
Examples
Write an email from bullet list
Code a snake game
Assist in a task
Search web
Ask anything
Model: meta-llama/Llama-2-70b-chat-hf · Generated content may be inaccurate or false.

OPEN.AI PLAYGROUND



TYPEN DES MASCHINELLEN LERNENS

Supervised Learning

Unsupervised Learning

Self-Supervised Learning

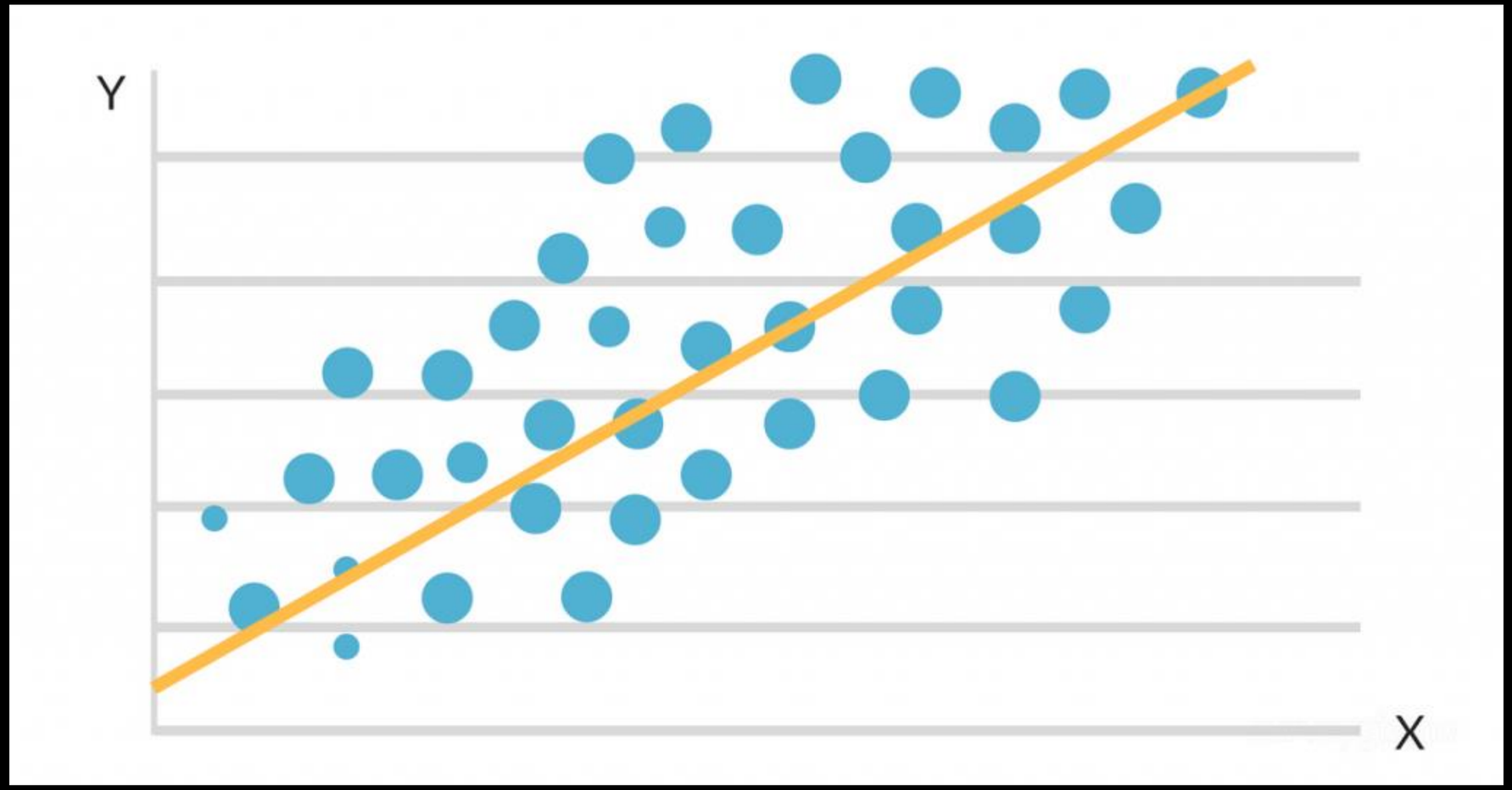
(Semi-Supervised Learning)

Reinforcement Learning

NVIDIA Blog: Supervised Vs. Unsupervised Learning. (2018, August 2). Retrieved May 4, 2021, from The Official NVIDIA Blog website: https://blogs.nvidia.com/blog/2018/08/02/supervised-unsupervised-learning/

Brownlee, J. (2019, November 10). 14 Different Types of Learning in Machine Learning. Retrieved May 4, 2021, from Machine Learning Mastery website: https://machinelearningmastery.com/types-of-learning-in-machine-learning/

REGRESSION: SUPERVISED LEARNING



AUFGABEN

- Trefft Euch einmal im Team, um mögliche zusätzlich zu erstellende Variablen zur Umsatzvorhersage zu diskutieren (ggf. auch ob es Daten gibt, die ihr zusätzlich verwenden könnt)

- Fügt im Verzeichnis "Data Import and Preparation" Eures Team Repositories erste zusätzliche Daten hinzu (z.B. zusätzliche heruntergeladene oder selbst erstellte Listen zu Feiertagen). Fügt ebenfalls ersten Code, der neue Variablen erstellt hinzu und erstellt den Programm-Code so, dass alle Daten in einem Datensatz zusammengeführt und gespeichert werden.

- Führt als Einführung zum Thema maschinelles Lernen die erste Woche des Kurses „AI for Everyone“ durch: https://www.coursera.org/learn/ai-for-everyone

coursera Explore What do you want to learn? For Enterprise Steffen Brandt

Browse > Business > Business Strategy

AI For Everyone

4.8 6.819 ratings • 2.028 reviews

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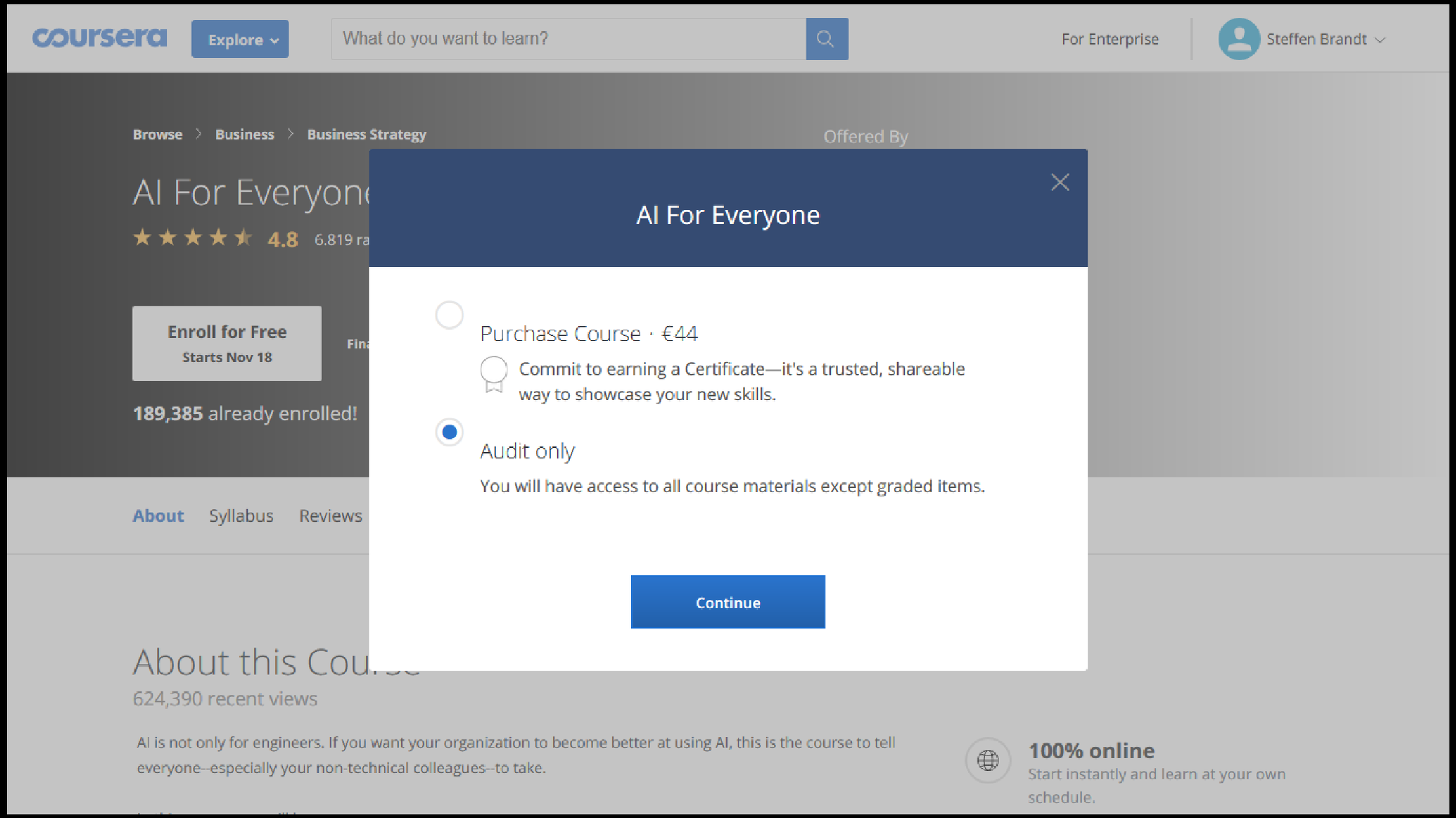
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Steffen Brandt
Browse
Business
Business Strategy
Offered By
AI For Everyone
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6.819 ra
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Starts Nov 18
189,385 already enrolled!
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